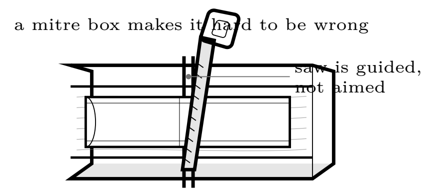
Craft 2: Cut Square

Shop skill — transferable to anything you ever build

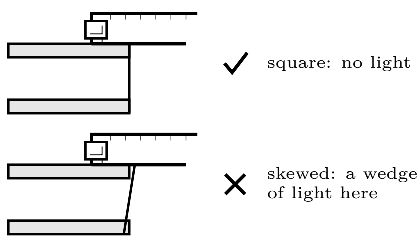
A joint can never be better than the cut that went into it

THE STANDARD

The cut face is **perpendicular to the pipe axis** within about one degree, and the whole face is flat. Held against a try square on a flat bench, you should not see light under the blade anywhere. On a 3 in pipe, one degree of skew is about 1.3 mm of gap on the far side — enough to matter, and easy to see once you know to look.



support the pipe on *both* sides of the cut



Why square matters — A socket joint is a precision taper. A skewed pipe end bottoms on one side while leaving a gap on the other, so the cement pools where it is not wanted and the weld area is smaller than it looks from outside. The joint appears finished and is not.

Use a guide — Freehand sawing a round object square is genuinely difficult and there is no reason to try. A mitre box or a tubing cutter turns a skill problem into an equipment problem.

Tubing cutter — Best result on pipe up to about 2 in — it cuts square by construction. Tighten a quarter turn per revolution, never more, or it ploughs and leaves a lip.

Fine teeth — On plastic, a fine-toothed saw cuts cleanly and a coarse one tears. Let the saw do the work; pushing hard makes the blade wander.

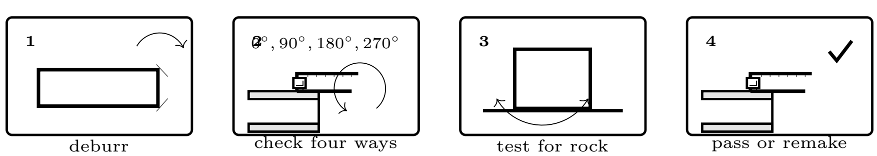
Support both sides — Pipe that drops as the cut finishes tears the last of the material and takes the face out of square in the final centimetre.

Cut on the waste side — Of the line, so the line survives on the piece you are keeping. *see the **Craft 1: Measure & Mark** sheet*

Method

1. Mark all the way round. A cut is only as square as its mark.
2. Set the pipe in the mitre box with the mark aligned to the slot, and hold it against the fence so it cannot roll.
3. Start the cut with the saw pulled backwards a couple of times to make a kerf. Do not start on the push.
4. Saw with long strokes and light pressure, letting the guide keep the angle. If thick pipe must be turned, stop, remove the saw, rotate the pipe, and secure it again before restarting. Never turn stock against a moving blade.
5. Support the offcut so it does not snap free.

Checking, and what to do if it is off



Test	What it tells you
Try square, light behind	The direct test. Rotate the square around the pipe to four positions — a cut can be square in one plane and skewed in the other.
Stand it on glass	A truly square end stands on a flat surface without rocking. Rock it gently; any wobble is skew.
Dry-fit and rotate	Push the pipe into its fitting and rotate it. If the gap at the shoulder changes as it turns, the end is not square.

Inspection decision

First remove only loose burrs; do not reshape the face. Ask: *Is there light at any of four square positions? Does the pipe rock on glass? Does the fitting shoulder gap change as the dry fit turns?*
Pass only if every answer is no and the finished length remains within 1 mm of the cut list. If any answer is yes, **re-cut from a new mark**. Do not sand the face square: it removes an unknown amount and can round the edge. If re-cutting would make the part undersize, **remake the part** from new stock. Keep the rejected piece only as a clearly marked practice coupon.

Check it yourself. Record the final proof: finished length ________ mm; light at 0° no / yes; 90° no / yes; 180° no / yes; 270° no / yes; rocking no / yes; rotating shoulder gap no / yes. Circle **PASS / RE-CUT / REMAKE**. Only PASS proceeds to a joint.

The habit worth taking away

Every trade has a version of this: use a guide rather than your eye, and then check against a known-good reference rather than against your own judgement of how it looks. A try square is a piece of steel that is more perpendicular than you are. That is not an insult, it is the entire reason it exists.